AeryxIQ

DRONE PACS AND AERIAL INSPECTION



ADMINISTRATION

Administrator Guide

People and roles, two-factor sign-in, the archive, reports and evidence, backup and restore, and the limits of this release.

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On-Premises Edition

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People and roles

Four roles, chosen so that the person who flies is not automatically the person who signs.

Role	Can
viewer	Read assets, missions, imagery and finalised reports. Changes nothing
pilot	Everything a viewer can, plus create missions, import imagery, record findings
manager	Everything a pilot can, plus finalise reports and manage the asset registry
admin	Everything, plus people, branding and organisation settings

Give people the lowest role that lets them do their job. A reviewing engineer who never flies is a manager, not an admin. A subcontracted operator is a pilot. A client who wants to see their own asset history is a viewer, and for a client who should see one study and nothing else, use a share link instead and give them no account at all.

Organisations

An organisation is the wall. Assets, missions, imagery, findings and reports belong to one, and nothing crosses. One installation can hold several: a parent company and its subsidiaries, or a service provider and the clients it keeps separate.

The sign-in page asks for a **Company ID** (the organisation's short ID, its slug), an **email** and a **password**. Because the Company ID is part of the sign-in, the same address can hold an account in more than one organisation without those accounts being related in any way. A sign-in pasted in the older combined form, `company-id:email`, into the Email box is split automatically.

```
powershell -ExecutionPolicy Bypass -File .\new-admin.ps1
./new-admin.sh
```

The same command creates an organisation that does not exist yet, and resets the password of an administrator who does. Everyone after the first is added from inside the application, under Admin.

Branding

Under Admin, set the organisation's display name, its brand colour and its logo. These are not decoration: they appear on the report PDFs your clients receive and on the public verification page an insurer or a records officer lands on. A service provider running several client organisations on one installation gets each one's reports in that client's own identity.

Removing the demo content

The sample content is loaded and removed from inside the application, under Admin, rather than by deleting rows. The removal is scoped to the seeded content and leaves anything you created alone. Do it before real flights go in, so that nobody has to work out later which buildings were real.

Two-factor sign-in

AeryxIQ implements time-based one-time passwords to RFC 6238. Any standard authenticator works: the phone apps, a password manager, a hardware token that does TOTP. Nothing is sent anywhere to set it up, which matters on an installation that has no outbound connection at all.

Enrolling

A user enrols themselves from their own account settings. They scan the code, type one generated six-digit code back to prove the clock agrees, and from then on sign-in asks for the code after the password.

Enrolment produces **recovery codes**. They are shown once. Only their hashes are stored, so nobody, including you, can recover them afterwards. Each works once. Tell people to put them somewhere that is not the phone holding the authenticator, because the failure this protects against is the phone.

Requiring it

An administrator can require two-factor sign-in for the whole organisation. Once set, a user without it is walked through enrolment at their next sign-in rather than being locked out.

It is worth requiring on any installation reachable from more than the one machine it runs on. A password is the only thing between an aerial archive and whoever is on the network otherwise.

When somebody is locked out

They use a recovery code. If those are gone too, an administrator disables two-factor for that user, who then re-enrols. Confirm who you are talking to by some means other than the email address first; an email address is exactly what an attacker in this position would have.

What else protects sign-in

Passwords are hashed with Argon2. Repeated failures lock an account for a period rather than allowing unlimited guessing. Sessions are signed with the key in `.env`; rotating it signs everybody out at once, which is the right first move if you believe a session has been taken.

The archive

Assets

An asset is the thing you keep coming back to. 62 types are built in, spanning energy, structures, land, and public safety work. The type is not a label: it selects the component vocabulary used when a finding is recorded, so a finding on a building can sit on a parapet, a coping or a spandrel, while one on a turbine sits on a blade root or a leading edge.

Choosing the type carelessly costs you later, when the vocabulary on the screen does not match the asset in front of the reader.

Re-inspection intervals

Each asset carries one, and it is the field that makes the product do something a folder of photographs cannot: tell you what is due. Buildings under a facade ordinance are typically on a five-year cycle, mandated bridge inspections on twenty-four months, most else annual. Set it when the asset is registered. An asset without one never appears on the Flight Plan, which is how an asset quietly stops being inspected.

Missions

A mission is one flight over one asset on one day. 46 types are built in. The type drives what the reading room expects: a thermal inspection and a volumetric survey are not read the same way.

Within a mission, imagery is organised into media sets, of nine kinds: photo, video, thermal, multispectral, orthomosaic, elevation model, point cloud, document and other. One flight commonly produces several.

Import

Copy the card off the aircraft and drop the folder on the mission. AeryxIQ reads each frame's EXIF capture time and GPS position. Where more than `FLIGHT_GAP_MINUTES` separates consecutive frames, what follows is filed as a separate mission. A card holding three sorties arrives as three studies.

Frames without GPS still import; they simply do not appear on the track. Frames without a capture time cannot be ordered or split, which is worth knowing before somebody strips metadata in an editor on the way in.

Capture devices

Devices are issued credentials of their own, listed, and revoked. A tablet in a van that is no longer in the fleet is revoked from one screen, without touching the account of the person who was using it.

Reading, grading, reporting

The reading room

Filmstrip down the side, the current frame in the middle, the flight track on the map. Drag a box around a defect to create a finding, which carries what it is, where on the asset it sits, and how bad it is.

Grading

Grade	Means	In practice
G1	Minor	Noted, no action
G2	Low	Monitor at the next cycle
G3	Moderate	Plan a repair
G4	Serious	Repair before the next cycle
G5	Critical	Act now

72 finding categories span structural, electrical, thermal, environmental, agricultural and public-safety work. Agree internally what each grade means for your assets and write it down. Consistency between two readers is worth more than precision in either of them, because the value of the archive is the comparison across years.

Reading against last year

Once an asset has more than one study, this year's imagery is read against the previous one. That is where the product earns its place: a crack that is present becomes a crack that is longer, and a determination stops being a judgement about a photograph and becomes a judgement about a trend.

Facade ordinance work

For buildings under a facade ordinance, classification follows the statutory vocabulary: **SAFE**, **SWARMP** (safe with a repair and maintenance programme), and **UNSAFE**. Where a finding sits is what moves it: a falling-hazard defect over a public sidewalk is not the same finding as the same defect over a roof.

> **THE DETERMINATION IS NOT THE SOFTWARE'S**
>
> A filing is made by a qualified inspector, from a physical inspection, under their own licence. AeryxIQ organises the evidence and proposes a classification from the findings recorded in it. It holds no certification and it signs nothing.

Reports

A report is drafted, then finalised. Finalising produces the PDF: the condition summary, the observation log, the annotated key images and a certification block in the organisation's own identity. Finalising is a manager's act, which is why pilot and manager are separate roles.

Sharing a study

A share link opens one mission in any browser, with no account and no install. It is scoped to that mission, it expires, and nothing else in the archive is reachable from it. That is what to give a client or a loss adjuster, rather than an account.

Verification

Finalised reports carry a verification link. Anyone who follows it, with no sign-in, gets every piece of evidence behind the report hashed again on the spot and compared with what was recorded when it was finalised. A court clerk, an insurer or a records officer can establish for themselves that the images behind a report are the images that were there on the day, without being given access to anything else.

For those links to work for somebody outside, `APP_HOST` must be an address they can actually reach. On a closed installation they will not resolve from outside the building, which is usually the intention and occasionally a surprise.

Closing the loop

A graded finding becomes a **work order**, which is tracked to completion and read back against the asset's history, so that next year's flight shows whether the repair happened. **Alerts** raise critical findings and pilot credential expiries, and are acknowledged by a person rather than simply disappearing. **Capital planning** turns graded findings across the portfolio into a costed plan, exported as a spreadsheet for whoever holds the budget.

Backup and restore

The part of this guide that will matter most, on a day you will not choose.

Taking one

```
powershell -ExecutionPolicy Bypass -File .\backup.ps1 -To D:\Backups
./backup.sh /backups/aeryxiq
```

Three files, timestamped, and all three are needed:

File	Holds
`aeryxiq-db-*.sql`	Assets, missions, findings, reports, users
`aeryxiq-media-*.tar.gz`	Every frame
`aeryxiq-env-*.txt`	The database password and the token signing key

The database without the imagery is an index of files that are gone. The imagery without the database is a pile of photographs with nothing saying what they are. Keep them together, and keep them somewhere that is not the machine being backed up.

THE THIRD FILE IS A SECRET

`aeryxiq-env-*.txt` holds the credentials to the archive. Wherever backups go deserves the same protection as the imagery.

Scheduling it

Windows, as a daily Task Scheduler task:

```
powershell -ExecutionPolicy Bypass -File C:\AeryxIQ\aeryxiq-1.0.1\backup.ps1 -To
D:\Backups
```

Linux, in cron:

```
0 2 * * * /opt/aeryxiq/aeryxiq-1.0.1/backup.sh /backups/aeryxiq
```

The script exits non-zero when the dump fails or comes back empty, so a scheduler that reports failures will tell you. One that does not report failures is not a backup system.

Restoring

On a clean installation, with the stack started once so the volumes exist:

```
docker compose down
cp aeryxiq-env-20261028-020000.txt .env
docker compose up -d postgres

docker compose exec -T postgres psql -U aeryxiq -d postgres \
  -c "DROP DATABASE IF EXISTS aeryxiq;" -c "CREATE DATABASE aeryxiq OWNER aeryxiq;"
docker compose exec -T postgres psql -U aeryxiq aeryxiq < aeryxiq-db-20261028-020000.sql

docker run --rm -v aeryxiq_onprem_media_data:/media_data \
  -v "$PWD":/in alpine tar xzf /in/aeryxiq-media-20261028-020000.tar.gz -C /media_data

docker compose up -d
```

Then sign in and open a mission with imagery in it. A restore that has not been looked at is not a restore.

> **DO THIS ONCE, NOW**
>
> Restore onto a spare machine while nothing is at stake. It takes an hour and it is the only way to know whether the backups you are taking are the backups you think you are taking. Every organisation that discovers otherwise discovers it on the worst possible day.

Running it day to day

Health

```
powershell -ExecutionPolicy Bypass -File .\health.ps1
./health.sh
```

Reports the containers, whether the application answers, whether the database is ready, and how much disk the volumes have taken. Disk is the one to watch. A full disk presents as general slowness for some time before it presents as an error, and by then imports may have failed quietly.

Logs

```
docker compose logs -f backend
docker compose logs --since 1h caddy
```

Security posture

What the product does:

- Encrypts traffic, including on the machine itself
- Hashes passwords with Argon2, and locks accounts after repeated failures
- Offers RFC 6238 two-factor sign-in, which an organisation can require
- Keeps organisations apart, which is tested rather than asserted
- Scopes share links to one mission, with an expiry
- Recomputes evidence hashes on demand for anyone holding a verification link
- Refuses to start on a weak token signing key
- Accepts browser calls only from its own origin
- Leaves unauthenticated organisation self-provisioning switched off
- Contacts nothing: no licence server, no telemetry

What it leaves to you:

- The disk is not encrypted by AeryxIQ. Use the operating system's own encryption on a machine that can be carried out of a building
- Backups are not encrypted either. They hold the imagery and the keys
- Who can reach the machine on the network is a firewall question
- Nobody is watching the logs unless somebody is watching the logs

What this release does not do

- One upload is capped at 2 GB. Larger folders go in as several; flight splitting still works across them

- Photogrammetry is off by default and is heavy when on
- Email, elevation lookup and transcription are unset, and blank is a working state
- The container images are built for 64-bit Intel and AMD machines. Apple Silicon runs them under emulation: correct, slower on import
- No automated upgrade path between major versions beyond changing the version and restarting, with a backup first

> **NO CERTIFICATION OF ANY KIND**
>
> AeryxIQ holds no engineering, surveying or inspection certification. Its automated assistance in detection, grading and drafting is assistance and nothing more. Every determination, grade, classification and signature remains the professional responsibility of the qualified person who makes it. Nothing in this guide, and nothing the software produces, changes a filing obligation or discharges a duty of care.

Support

Buy Support and **Upgrade to Cloud** are in the sidebar and on the sign-in page. Upgrade to Cloud moves the installation to AeryxIQ Cloud, which Amhage runs and backs up.

Support plan and professional installation: `aeryxiq.com/services/`
`michael.romansky@amhage.com`

Have the output of `health`, the version from the sign-in page, and the last fifty lines of the backend log ready.